CITS5508 Practice Exam — Chapter 5

Support Vector Machines

- This practice paper has **5 questions** for **60 marks**.
- Allow approximately **two minutes per mark** ($\approx$ 2 hours total).
- Only UWA-approved calculators permitted.
- Answers are in the separate file *Practice Exam - Chapter 5 (Answer Key).md*.

Question 1. (12 marks)

(a) Describe the **training procedure** of a linear SVM classifier. Your answer should explain: how the **margin** is defined, what the **support vectors** are, which training instances are relevant for defining the **decision boundary**, and what quantity is used as the **decision rule** for prediction. *(6 marks)*

(b) Explain what it means to **maximise the margin**, and why this is expressed as **minimising $\frac{1}{2}\|\mathbf{w}\|^2$** rather than maximising the margin directly. *(3 marks)*

(c) Why is it important to **scale the input features** before training an SVM? *(3 marks)*

Question 2. (12 marks)

(a) Explain the difference between **hard margin** and **soft margin** classification, and give **two** problems with hard margin classification. *(5 marks)*

(b) Describe the effect of the regularisation hyperparameter **C** on a linear SVM. If your SVM is **overfitting**, should you increase or decrease C? *(4 marks)*

(c) Can a `LinearSVC` output a **probability** for its predictions? What about a confidence **score**? Explain. *(3 marks)*

Question 3. (12 marks)

(a) Explain what the **kernel trick** is and why it is useful. *(4 marks)*

(b) State the formula (or describe) the **Gaussian RBF kernel**, and explain the effect of increasing the hyperparameter **γ (gamma)** on the decision boundary. Is γ behaving like a regularisation parameter — if the model underfits, should you increase or decrease γ? *(4 marks)*

(c) According to **Mercer's theorem**, why can you use a kernel even without knowing the feature-space transformation φ? What is special about the φ corresponding to the RBF kernel? *(4 marks)*

Question 4. (12 marks)

(a) Compare the scikit-learn classes **`LinearSVC`**, **`SVC`**, and **`SGDClassifier`** for SVM classification, in terms of (i) computational/time complexity, (ii) support for the kernel trick, and (iii) suitability for very large or out-of-core datasets. *(6 marks)*

(b) You have a **linearly separable** dataset with 2 million instances and want a linear SVM. Which class would you choose and why? *(3 marks)*

(c) You have a small (~1,000 instance) dataset that is clearly **not** linearly separable. Which class and kernel would you try, and why? *(3 marks)*

Question 5. (12 marks)

(a) Explain how SVMs are adapted for **regression** (SVM regression). What does the hyperparameter **ε** control, and what does it mean for the model to be **ε-insensitive**? *(5 marks)*

(b) Write a short scikit-learn snippet that builds a pipeline scaling the features and then training an **RBF-kernel SVM classifier** with `C=10` and `gamma=0.5`. *(4 marks)*

(c) The SVM optimisation problem can be solved in its **primal** or **dual** form. State one reason you might solve the **dual** problem, and explain its connection to the kernel trick. *(3 marks)*